

§ 2.2.

1. Solution: Characteristic equation.

$$r^2 - 1 = 0$$

$$\Rightarrow r_1 = 1, r_2 = -1$$

Hence, $y_1(t) = e^t$ and $y_2(t) = e^{-t}$ are a fundamental set of solutions of this equation, so that

$$y(t) = c_1 e^t + c_2 e^{-t}$$

3. Solution: Characteristic equation:

$$r^2 - 3r + 1 = 0$$

$$\Rightarrow r = \frac{3 \pm \sqrt{9-4}}{2}$$

$$\Rightarrow r_1 = \frac{3}{2} + \frac{\sqrt{5}}{2}, r_2 = \frac{3}{2} - \frac{\sqrt{5}}{2}$$

So that

$$y(t) = c_1 e^{\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)t} + c_2 e^{\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right)t}$$

4. Solution: Characteristic equation:

$$3r^2 + 6r + 2 = 0$$

$$\Rightarrow r = \frac{-6 \pm \sqrt{6^2 - 24}}{6} = -1 \pm \frac{\sqrt{2}}{6}$$

$$\Rightarrow r_1 = -1 + \frac{\sqrt{2}}{6}, r_2 = -1 - \frac{\sqrt{2}}{6}$$

So that

$$y(t) = c_1 e^{(-1 + \frac{\sqrt{2}}{6})t} + c_2 e^{(-1 - \frac{\sqrt{2}}{6})t}$$

5. Solution: Characteristic equation.

$$r^2 - 3r - 4 = 0$$

$$\Rightarrow r_1 = 4, r_2 = -1$$

So that

$$y(t) = c_1 e^{4t} + c_2 e^{-t}$$

$$y(0) = 1 \Rightarrow c_1 + c_2 = 1$$

$$y'(0) = 0 \Rightarrow 4c_1 - c_2 = 0$$

$$\Rightarrow c_1 = \frac{1}{5}, c_2 = \frac{4}{5}$$

$$\Rightarrow y(t) = \frac{1}{5}e^{4t} + \frac{4}{5}e^{-t}$$

6. Solution: Characteristic equation

$$2r^2 + r - 10 = 0$$

$$\Rightarrow r_1 = 2, r_2 = -\frac{5}{2}$$

So that (since $t_0 = 1$)

$$y(t) = c_1 e^{r_1(t-t_0)} + c_2 e^{r_2(t-t_0)}$$

$$= c_1 e^{2(t-1)} + c_2 e^{-\frac{5}{2}(t-1)}$$

$$y(1) = 5 \Rightarrow c_1 + c_2 = 5$$

$$y'(1) = 2 \Rightarrow 2c_1 - \frac{5}{2}c_2 = 2$$

$$\Rightarrow c_1 = \frac{29}{9}, c_2 = \frac{16}{9}$$

$$\Rightarrow y(t) = \frac{29}{9}e^{2(t-1)} + \frac{16}{9}e^{-\frac{5}{2}(t-1)}$$

7. Solution: Characteristic equation:

$$5r^2 + 5r - 1 = 0$$

$$\Rightarrow r_1 = -\frac{1}{2} + \frac{3}{10}\sqrt{5}, r_2 = -\frac{1}{2} - \frac{3}{10}\sqrt{5}$$

So that

$$y(t) = c_1 e^{(-\frac{1}{2} + \frac{3}{10}\sqrt{5})t} + c_2 e^{(-\frac{1}{2} - \frac{3}{10}\sqrt{5})t}$$

$$y(0) = 0 \Rightarrow c_1 + c_2 = 0$$

$$y'(0) = 1 \Rightarrow (-\frac{1}{2} + \frac{3}{10}\sqrt{5})c_1 + (-\frac{1}{2} - \frac{3}{10}\sqrt{5})c_2 = 1$$

$$\Rightarrow c_1 = \frac{\sqrt{5}}{3}, c_2 = -\frac{\sqrt{5}}{3}$$

$$\Rightarrow y(t) = \frac{\sqrt{5}}{3} e^{(-\frac{1}{2} + \frac{3}{10}\sqrt{5})t} - \frac{\sqrt{5}}{3} e^{(-\frac{1}{2} - \frac{3}{10}\sqrt{5})t}$$

11. Solution: We guess $y = t^r$ (why?).

$$L[y] = t^2(t^r)'' + 5t(t^r)' - 5(t^r)$$

$$= (r(r-1) + 5r - 5)t^r$$

$$= (r^2 + 4r - 5)t^r = 0$$

$$\Rightarrow r^2 + 4r - 5 = 0$$

$$\Rightarrow r_1 = -5, r_2 = 1$$

$$\Rightarrow y(t) = c_1 t^{-5} + c_2 t$$